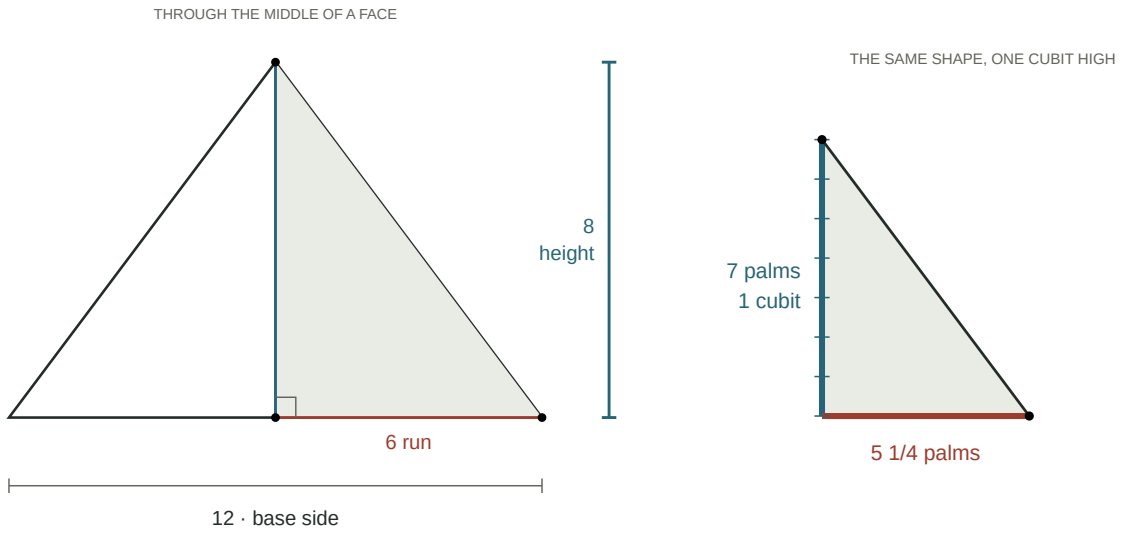
Problem 59: When the statement and working disagree

The literal edition states height 12 and base side 8, then works with 8 to make 6. Read both sets of dimensions.

Selected step 4: **Notice the second discrepancy**. Selected dimensions: used in the working (side 12, height 8).



A modern section at equal scales: base 12, height 8. The horizontal measure is **5 1/4 palms per vertical cubit**. One cubit is seven palms; one palm is four fingers.

Chace's literal edition retains a result of five palms and two fingers and marks "2" as a mistake for "1." Peet's translation prints the corrected one finger. Neither result agrees with height 12 and base side 8 as written.

1. ### Keep the written statement visible

   Height 12 and base side 8 give a half-base of 4. That triangle has a seked of 2 1/3 palms. Use the reading switch to compare it with the dimensions implied by the working.

   > as written: 7 × 4/12 = 7/3 palms

2. ### Read what the arithmetic actually does

   The working asks for the fraction of 8 that makes 6. A half and a quarter do that. This matches a height of 8 and a base side of 12, reversing the stated dimensions.

   > 8 × (1/2 + 1/4) = 6

3. ### Convert the worked fraction

   Multiplying three quarters by seven gives 5 1/4 palms: five palms and one finger. The unit-fraction rows support this number.

   > 7 × (1/2 + 1/4) = 21/4 palms

4. ### Notice the second discrepancy

   The literal edition prints five palms and two fingers, or 5 1/2 palms, and proposes one finger in a footnote. The working's 5 1/4 and the printed result's 5 1/2 stay separate here.

   > worked: 21/4; printed result: 11/2 palms

5. **Compare the editions, not just the answers**

   Peet prints the corrected five palms and one finger. Chace’s literal translation preserves two and footnotes the correction. The disagreement is visible in the included printed pages.

   > statement ≠ working ≠ literal result

Selected reckoning rows

1/2 + 1/4: multiplier 3/4 × 8 = 6. Target 6. Target reached.

This table follows the arithmetic’s 8 and 6, even when the drawing is set to the written statement. Half plus quarter makes three quarters. The third and two-thirds rows are modern comparisons.

Exact quantities for this reading

Quantity	Exact value
Base side	12
Vertical height	8
Half-base	6
Horizontal cubits per vertical cubit	3/4
Palms per vertical cubit	21/4
Fingers per vertical cubit	21

The experiment controls were saved at base 12, height 8, section corner. The lesson drawings above retain the selected source dimensions.

Words in the printed edition

mr · pyramid

The noun names the subject of problems 56–59B. The source drawing includes a low base line; the geometry in this app is a modern section.

mḥ · cubit

The arm measure appears when the calculation reduces the large triangle to a one-cubit rise. These exercises explicitly use a cubit of seven palms.

šsp · palm

Seven palms make the cubit used here. One finger is a quarter of a palm. The horizontal result can therefore be stated in palms and fingers.

iwn · pillar; a structure of uncertain form here

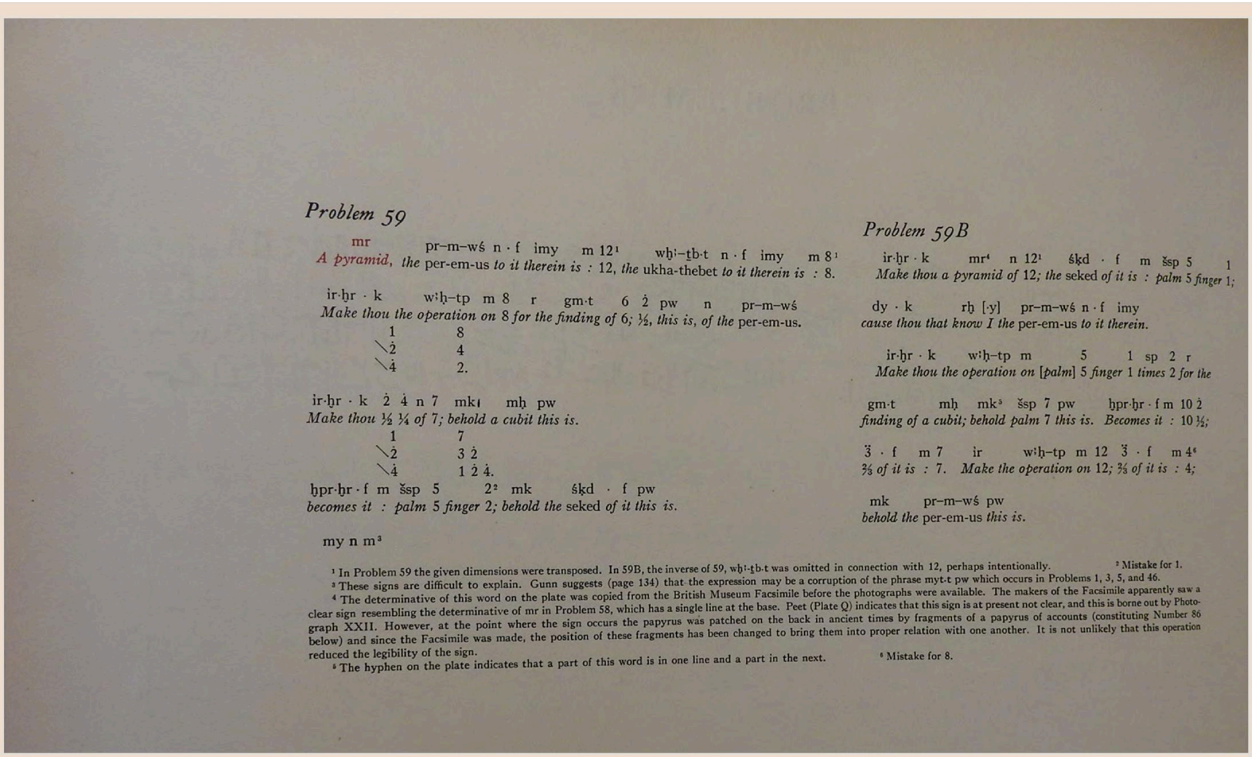
The word replaces “pyramid” in problem 60. A vocabulary gloss alone does not establish whether this particular figure is a cone, pyramid or another solid.

These are consonantal transliterations, not recovered pronunciations. The drawings are modern explanations; they do not reconstruct a named monument or its construction method. The face-slope interpretation follows the cited editions.

The complete source pages

Chace, Bull and Manning, *The Rhind Mathematical Papyrus*, volume II (1929). The transcription, literal translation and the two papyrus photographs are separate kinds of evidence. The plates reverse the transliteration's letters and digits for right-to-left reading; the facing literal translation reads left to right. Peet's 1923 p. 100 is also included for comparison.

Problem 59 & 59B · literal translation. A dot above the first digit denotes a unit fraction in this edition; a double dot above 3 denotes two thirds. The notes explain proposed corrections. Scanned edition.



Problem 59

mr pr–m–wś n · f imy m 12[1] wḫ[1]–ṯb·t n · f imy m 8[1]
A pyramid, the per-em-us *to it therein is : 12, the* ukha-thebet *to it therein is : 8.*

ir·ḫr · k wꜥḫ–tp m 8 r gm·t 6 2̇ pw n pr–m–wś
Make thou the operation on 8 for the finding of 6; ½, this is, of the per-em-us.

1 8
\2̇ 4
\4̇ 2.

ir·ḫr · k 2̇ 4̇ n 7 mk[1] mḫ pw
Make thou ½ ¼ of 7; behold a cubit this is.

1 7
\2̇ 3 2̇
\4̇ 1 2̇ 4̇.

ḫpr·ḫr · f m šsp 5 2[2] mk śḳd · f pw
becomes it : palm 5 finger 2; behold the seked *of it this is.*

my n m[3]

Problem 59B

ir·ḫr · k mr[4] n 12[1] śḳd · f m šsp 5 1
Make thou a pyramid of 12; the seked *of it is : palm 5 finger 1;*

dy · k rḏ [·y] pr–m–wś n · f imy
cause thou that know I the per-em-us *to it therein.*

ir·ḫr · k wꜥḫ–tp m 5 1 sp 2 r
Make thou the operation on [palm] 5 finger 1 times 2 for the

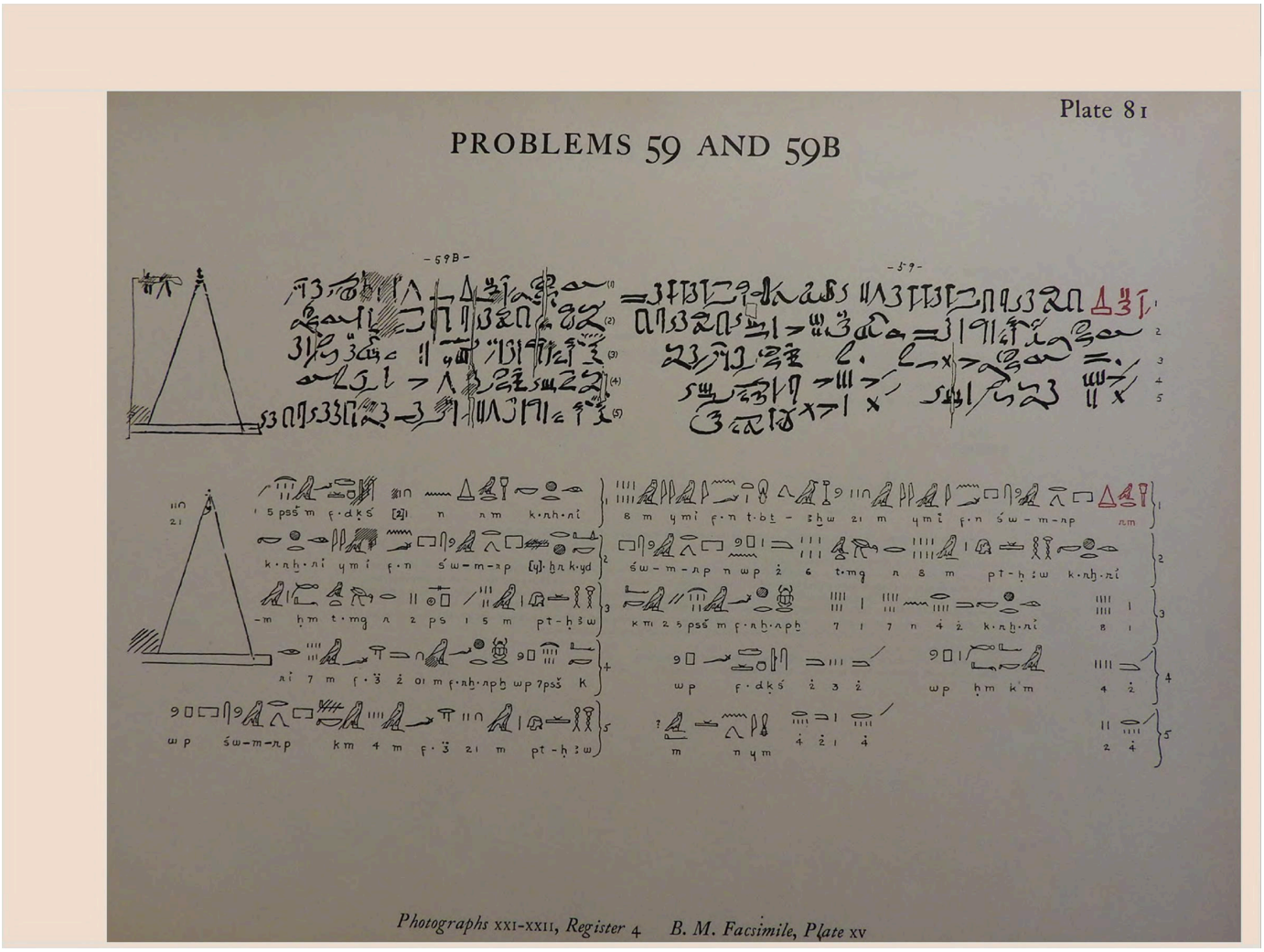
gm·t mḫ mk[5] šsp 7 pw ḫpr·ḫr · f m 10 2̇
finding of a cubit; behold palm 7 this is. Becomes it : 10 ½;

3̈ · f m 7 ir wꜥḫ–tp m 12 3̈ · f m 4[6]
⅔ of it is : 7. Make the operation on 12; ⅔ of it is : 4;

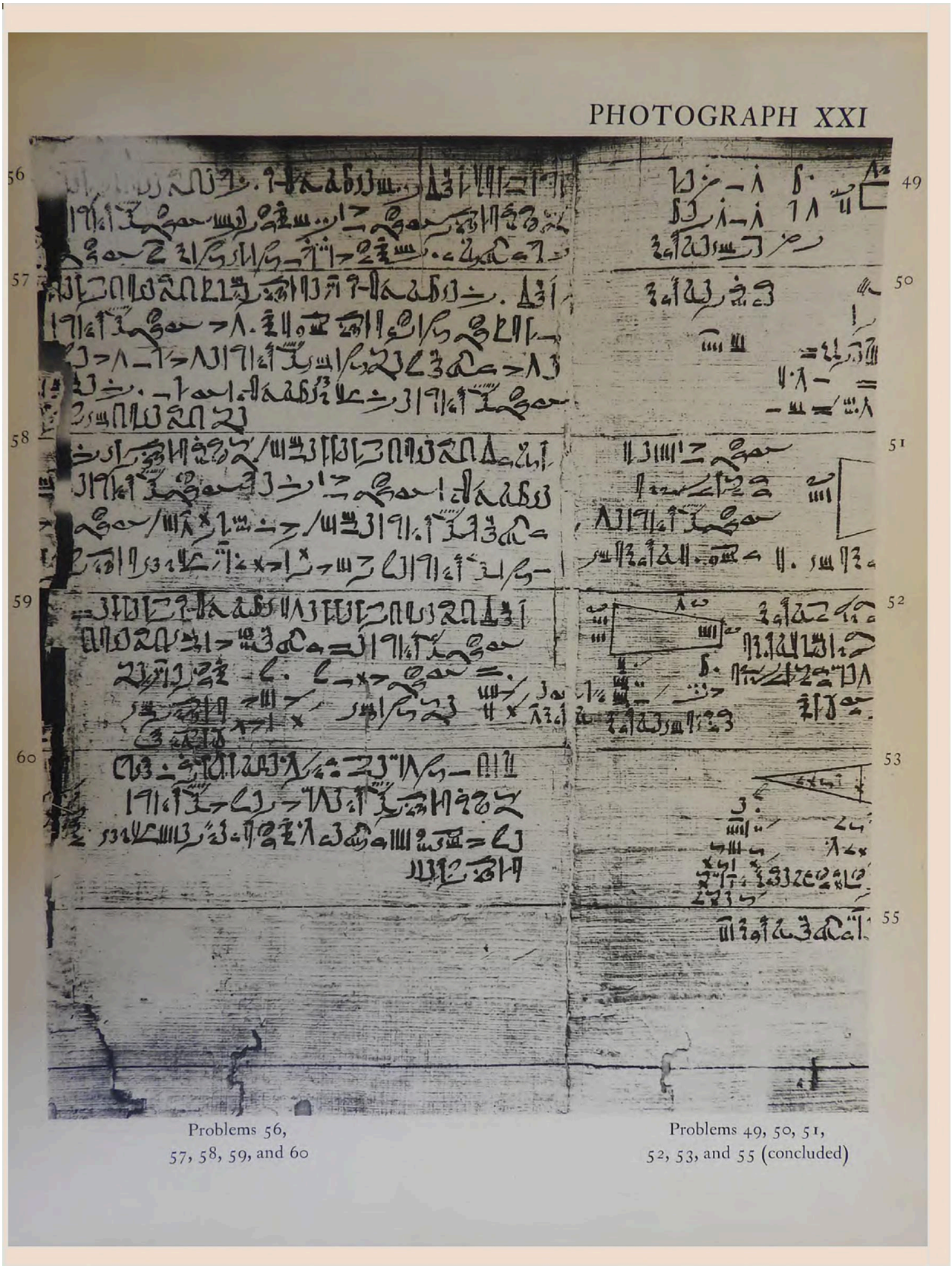
mk pr–m–wś pw
behold the per-em-us *this is.*

[1] In Problem 59 the given dimensions were transposed. In 59B, the inverse of 59, wḫ[1]-ṯb·t was omitted in connection with 12, perhaps intentionally. [2] Mistake for 1.
[3] These signs are difficult to explain. Gunn suggests (page 134) that the expression may be a corruption of the phrase myt·t pw which occurs in Problems 1, 3, 5, and 46.
[4] The determinative of this word on the plate was copied from the British Museum Facsimile before the photographs were available. The makers of the Facsimile apparently saw a clear sign resembling the determinative of mr in Problem 58, which has a single line at the base. Peet (Plate Q) indicates that this sign is at present not clear, and this is borne out by Photograph XXII. However, at the point where the sign occurs the papyrus was patched on the back in ancient times by fragments of a papyrus of accounts (constituting Number 86 below) and since the Facsimile was made, the position of these fragments has been changed to bring them into proper relation with one another. It is not unlikely that this operation reduced the legibility of the sign.
[5] The hyphen on the plate indicates that a part of this word is in one line and a part in the next. [6] Mistake for 8.

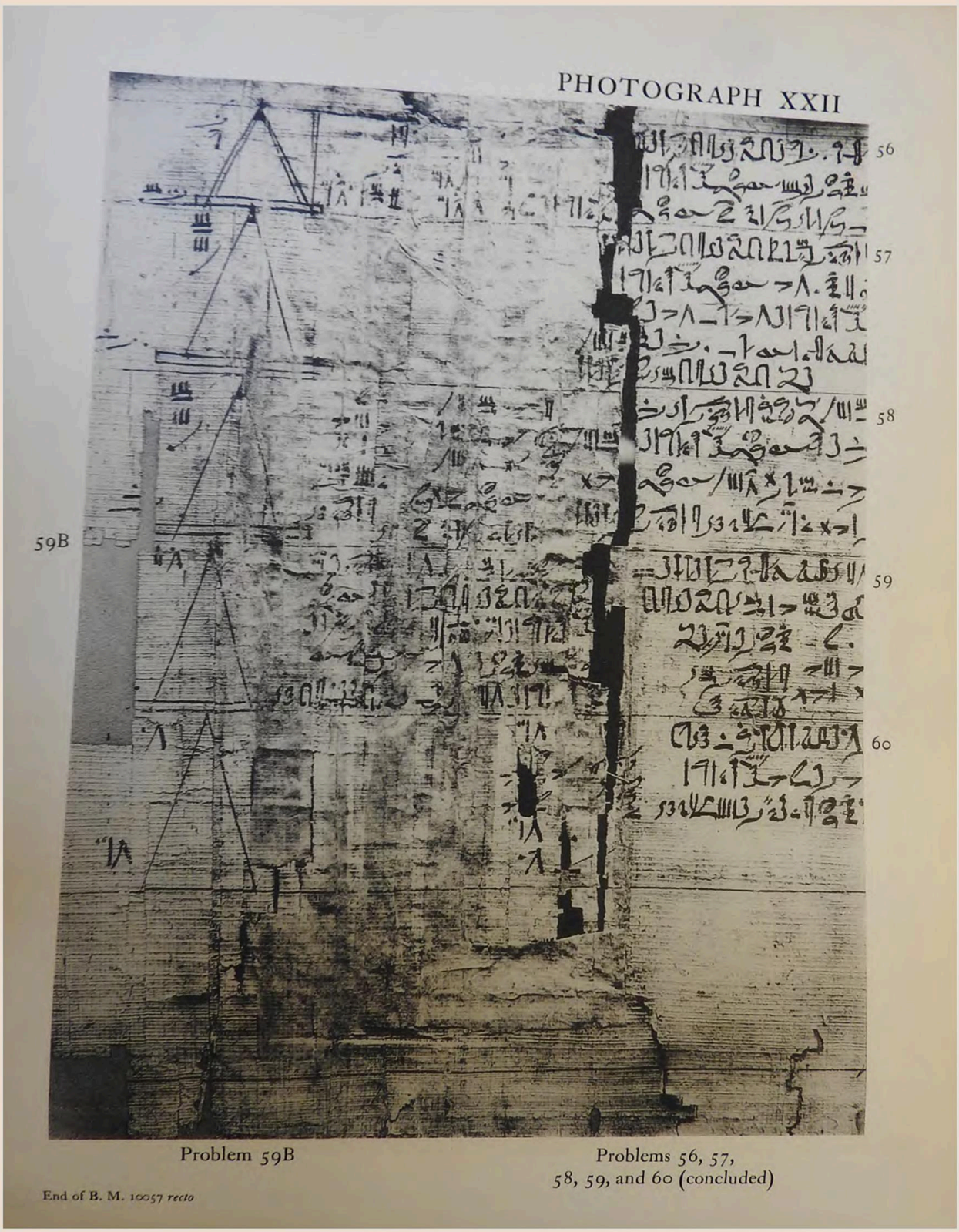
Problem 59 & 59B · transcription, plate 81. Copied hieratic above, hieroglyphic transcription below. The edition's transliteration runs right to left, with letters and digits reversed. Compare the facing literal translation for left-to-right reading. Scanned edition.



Photograph XXI · whole papyrus portion. Problem 59 occupies register 4, counted from the top: its beginning is at the left of photograph XXI, and its continuation at the right of XXII. The photographs overlap. The printed problem numbers were added by editors. Scanned edition.



Photograph XXII · adjoining papyrus portion. Problem 59 occupies register 4, counted from the top: its beginning is at the left of photograph XXI, and its continuation at the right of XXII. The photographs overlap. The printed problem numbers were added by editors. Scanned edition.



Peet (1923), p. 100, from the 1970 facsimile reprint: compare the printed corrections in problems 59 and 59B. Scanned edition.

No. 59. No. 59. (Pl. Q.)

"A pyramid the vertical height (*sic*) whereof is 12 and the side (*sic*) 8. ⟨Find its batter.⟩

You are to reckon with 8 to find 6, for this is half the height.

1	8
$\frac{1}{2}$	4
$\frac{1}{4}$	2

You are to take a half and a quarter of 7, for this is 1 cubit.

1	7
$\frac{1}{2}$	$3\frac{1}{2}$
$\frac{1}{4}$	$1\frac{1}{2}+\frac{1}{4}$

It comes to 5 palms 1 finger. Behold this is its batter.

What . . . . . ?"

No. 59b. No. 59B. (Pl. Q.)

"You are to reckon a pyramid of 12 (*sic*), whose batter is 5 palms 1 finger. Let ⟨me⟩ know the height thereof.

You are to reckon with $5\frac{1}{4}$ doubled to find 1 cubit, which is 7 spans: it comes to $10\frac{1}{2}$. Two-thirds of it is 7.

Reckon with 12: two-thirds of it is 4 (read 8). Behold this is the height."

Nos. 59 and 59B are two quite distinct problems, the second of which is the reverse of the first. Unfortunately the scribe did not realize this, and has made No. 59B appear as part of the working of No. 59 by introducing it by the word *ỉr·ḫr·k* and failing to use red ink for the opening word or words. The verbal form *šḏm·ḫr·k* is never used in this papyrus to introduce a problem. In the original from which our scribe copied the problem was probably, like its fellows, introduced directly by *mr* "a pyramid." It is possible that this was in black instead of being, as it should be, in red. We know enough of our scribe's intelligence to assert that this would be quite sufficient to conceal from him the fact that a fresh problem had begun.

In No. 59 there is an unfortunate error in the setting out, for the side and the height have been transposed. In order to give a batter of 5 palms 1 finger it is the height which must be 8 and the side 12. Otherwise the batter would be only $2\frac{1}{3}$ spans.

The phrase which ends the problem is a mystery. It may be corrupt. Is it connected with the scribe's error in running the next problem on to this?

In No. 59B we are given the side and batter and asked to find the height. After the words "a pyramid of 12" we expect *m wḏ;-ṯbt·f* "in its side"; but it is quite possible that the Egyptian is sufficient as it stands, and that "a pyramid of 12" was the technical expression for "a pyramid built on a base 12 square."

No. 60. No. 60. (Pl. R.)

"A cone (?) of 15 cubits in its base and 30 in its height. Let me know its batter.

Reckon with 15: its half is $7\frac{1}{2}$.

Multiply $7\frac{1}{2}$ four times (*sic*) to find 30.

The result is 4. This is the batter thereof.

Working:

1	15
$\frac{1}{2}$	$7\frac{1}{2}$
1	$7\frac{1}{2}$
2	15
4	30 "

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